

AISSAOUI Ismail

AI & Data Science Engineer

i.aissaoui@esi-sba.dz | +213 660 70 77 96 | Chlef, Algeria
[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI & Data Science Engineer specializing in advanced machine learning architectures, including Mamba-SSM, PINNs, and GNNs. Expertise in architecting ultra-low latency edge AI solutions and scalable cloud models for industrial and medical sectors. Skilled in high-performance computing (C++, Next.js) and modern web-based 3D visualization.

TECHNICAL SKILLS

Languages: Python, C++, Next.js, TypeScript, JavaScript, SQL
AI/ML: PyTorch, TensorFlow, Mamba, PINNs, GNNs, Transformers, DeepONet, KAN
3D/Web: React, Next.js, Three.js, React Three Fiber, FastAPI, TailwindCSS
Data/MLOps: MLflow, PyArrow, Docker, Linux, Git, PostgreSQL, Parquet
Optimization: Post-Training Quantization (INT8), RL, Genetic Algorithms

ACADEMIC THESES

Engineering Thesis: A Dual-Track Cloud-Edge Digital Twin for Gas Turbine Predictive Maintenance 2026
Sonatrach (Industrial Surveillance Division)

- Architecture:** Designed a 3-tier framework decoupling safety-critical Edge models from Cloud analytics.
- Optimization:** Engineered **TCNEdge** and compiled 1D Dilated Convolutions to INT8 ONNX C++, achieving 0.049ms latency.
- Robustness:** Developed **MambaTS** with missingness-aware gating for uninterrupted inference during sensor loss.
- Physics-Informed:** Implemented **KAN-PCNN** and **ST-KDFormer** to ensure thermodynamic consistency via PyTorch Autograd.

Master's Thesis: Hemodynamic Digital Twins: A Physics-Informed AI Approach for Critical Care Prognostics 2026
Higher School of Computer Science (ESI-SBA)

- Architecture:** Engineered **Medical_MambaTS** for bedside inference and dual-branch **SST_KODA_MultiScale** for cloud forecasting using 100Hz telemetry.
- Modeling:** Implemented a 0D Windkessel **PI_DeepONet** to enforce cardiovascular fluid dynamics without cross-patient retraining.
- Efficiency:** Developed a zero-copy pipeline with PyArrow Parquet to eliminate GPU starvation bottlenecks.

PROFESSIONAL PROJECTS

Aura — Enterprise AI Trading Platform 2025

- Forecasting:** Built a hybrid system using LSTMs for high-fidelity time-series predictions.
- Policy:** Implemented Deep Q-Learning (RL) to optimize long-term trading strategies.
- Deployment:** Engineered a high-throughput FastAPI backend with WebSockets and a Next.js UI.

Career Connect — GNN-based Recruiting System 2024 – 2025

- Relational Learning:** Developed a scalable recruiting engine using GNNs to map candidate-job relationships.
- Processing:** Utilized PyTorch Geometric and SBERT for deep semantic embeddings.
- Architecture:** Deployed via FastAPI and PostgreSQL, significantly improving match precision.

EDUCATION

National Higher School of Computer Science (ESI-SBA) 2021 – 2026
State Engineer and Master's Degree in Artificial Intelligence & Data Science
Defended June 28, 2026

CERTIFICATIONS & LANGUAGES

- DeepLearning.AI:** Machine Learning Specialization, Deep Learning Specialization, Natural Language Processing Specialization
- Zero To Mastery (ZTM) Academy:** Complete PyTorch: Deep Learning and Artificial Intelligence, TensorFlow Developer Certificate 2024
- Languages:** Arabic (Native), English (Advanced), French (Professional)